

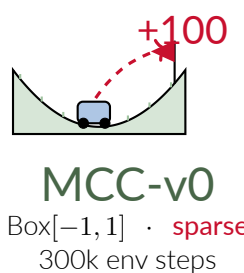
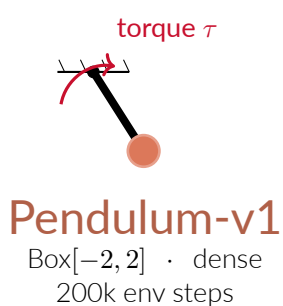
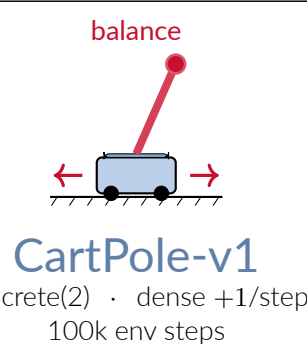
A Comparative Study of Deep RL Algorithms: DQN, PPO, SAC, TD3

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1. Four classic-control environments



2. Setup and taxonomy

We re-implement four canonical deep RL algorithms from scratch in PyTorch (no **stable-baselines3**) and benchmark them with 3 seeds each, 95% bootstrap CIs over seeds, and four pre-registered ablations on four Gymnasium [5] classic-control environments.

	on-policy	off-policy
value-based	—	DQN
policy-based	PPO	SAC, TD3

Networks. (64, 64) MLPs for DQN, PPO; (256, 256) for SAC, TD3 (canonical paper widths). **Hardware.** RTX 3070 + EPFL gnoto V100.

3. Algorithms

DQN [1]

value-based, off-policy, discrete:

$$\mathcal{L} = \mathbb{E}[\ell_{\text{Hub}}(Q_{\theta} - r - \gamma(1-d) \max_{i'} Q_{\theta})]$$

ϵ -greedy, target sync every τ steps, Huber loss.

PPO [2]

policy-based, on-policy:

$$\mathcal{L} = -\mathbb{E}[\min(r_t \hat{A}_t, \text{clip}(r_t, 1 \pm \epsilon) \hat{A}_t)]$$

GAE advantages, clipped surrogate as a soft trust region.

SAC [3]

off-policy, max-entropy:

$$J = \mathbb{E}[\sum_t \gamma^t (r_t + \alpha \mathcal{H}[\pi(\cdot|s_t)])]$$

Squashed-Gaussian actor, auto α via $\log \alpha$, twin Q.

TD3 [4]

off-policy, deterministic:

$$y = r + \gamma(1-d) \min_i Q_{\phi_i}(s', \text{clip}(\pi_{\theta} + \text{clip}(\tilde{\epsilon}, \pm 0.5), a_{\min}, a_{\max}))$$

Twin critics, target smoothing $\tilde{\sigma}=0.2$; two clips required.

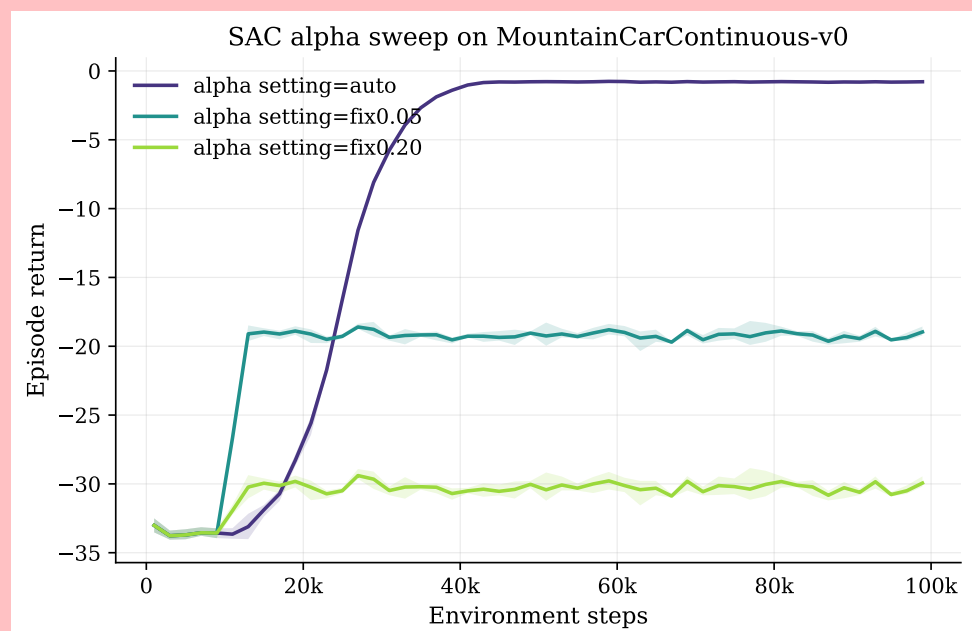
4. Implementation details that mattered

- **DQN:** target-sync sweep on CartPole gives a **U-shape** ($\tau \in \{100, 1000, 5000\}$). The canonical $\tau=1000$ sits at the edge of stability, which explains the seed-bimodality.
- **PPO:** at Pendulum truncation we fold $\gamma V(s_{t+1})$ into the reward so GAE accounts for the value of the truncated state; raw **done** flags bias returns by ~ 30 .
- **SAC:** the $\tanh$ -Jacobian log-prob correction $-\sum_i \log(1 - \tanh^2 u_i)$ is mandatory; without it the entropy term is silently wrong and training never moves off the warmup floor (verified: return stuck at -1200 on Pendulum without it).
- **TD3:** the smoothed target action needs two clips, on the noise (± 0.5) and on the action range. Skipping the outer clip sticks the actor on the boundary.
- **All four:** bootstrap mask uses **terminated** only, not **terminated or truncated**.

5. Pre-registered ablations: turning the canonical knobs

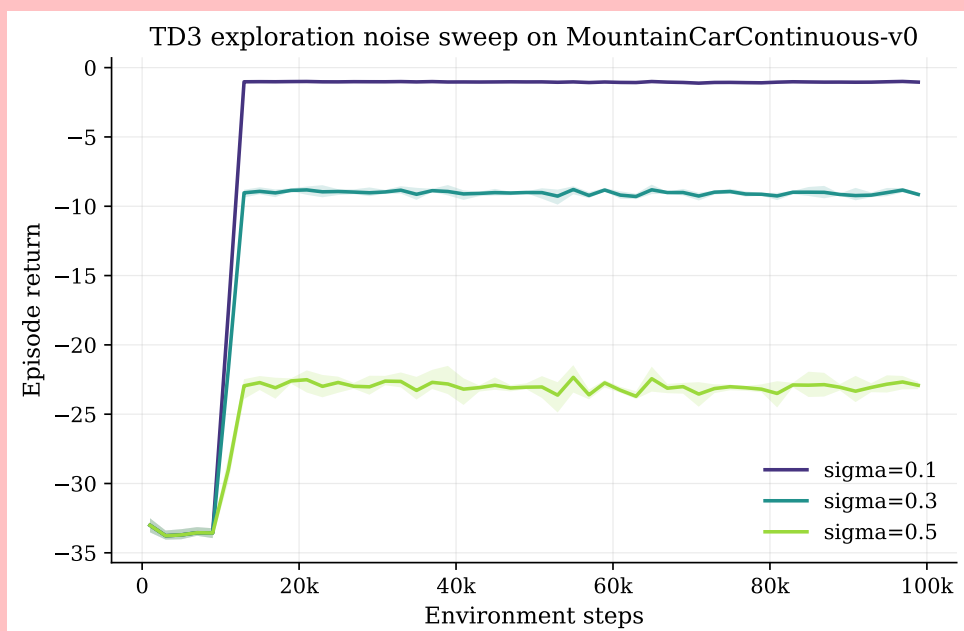
Four pre-registered sweeps, one per algorithm, 3 seeds each.

Headline (MCC, sparse reward): turning the off-policy exploration knobs (SAC α , TD3 σ) does not rescue them.



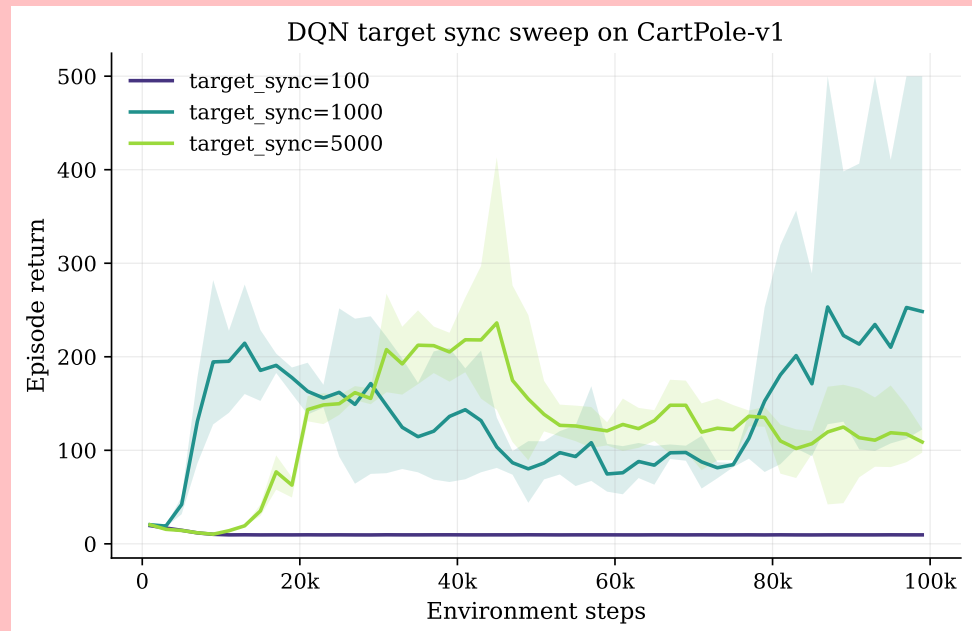
SAC α sweep on MCC.

Auto, 0.05, 0.20: three failure modes, 0 goals / 9 seeds.



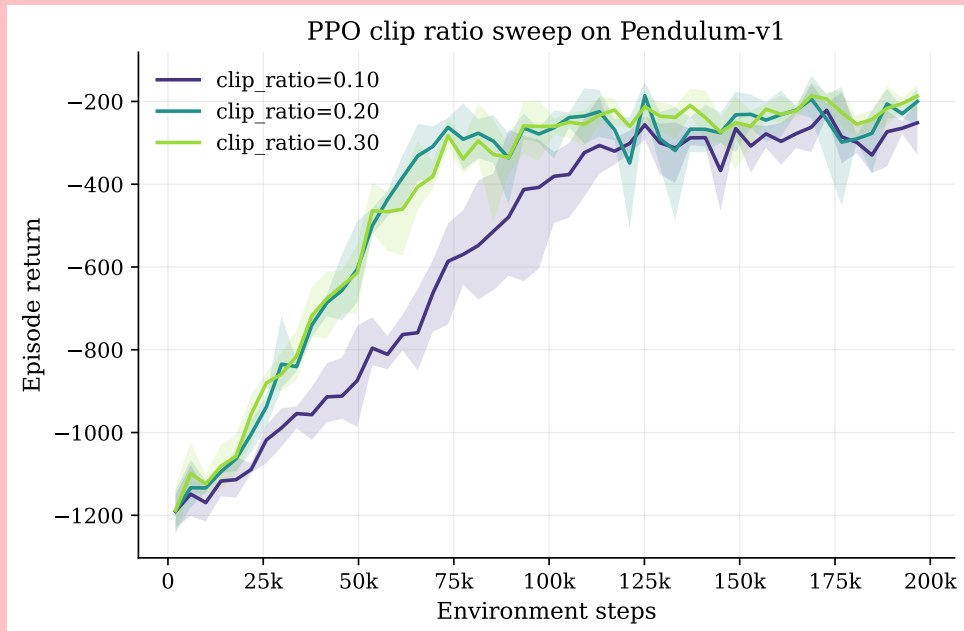
TD3 σ sweep on MCC.

$\sigma \in \{0.1, 0.3, 0.5\}$: more noise is worse.



DQN target-sync, CartPole.

U-shape: $\tau=1000$ at the edge of stability.



PPO clip ratio, Pendulum.

$\epsilon=0.1$ too conservative; 0.2 & 0.3 tied.

Why off-policy continuous methods fail on MCC. Entropy regularization (SAC) and additive Gaussian noise (TD3) drive uniform coverage of the *local* action set, not directed exploration in state space. PPO's on-policy rollouts occasionally produce a long correlated action sequence that crosses the hill; off-policy replay samples are decorrelated and cannot. Structured exploration variants (e.g. gSDE) have been shown to close this gap [6].

6. Limitations & open questions

- **More seeds.** 3 per (algo, env); more would tighten CIs at super-linear cost. Bootstrap CIs are wide enough to keep claims honest.
- **Network width sweep.** Widths fixed at canonical defaults: (64, 64) for DQN/PPO, (256, 256) for SAC/TD3. The actor-parameter gap is a width choice, not an algorithmic property.
- **Second sparse benchmark.** Generalizing the MCC headline needs e.g. **Reacher** with a binary reward or a sparsified **Pendulum**.
- **Directed exploration.** Add gSDE [6], RND, or NoisyNet on top of SAC/TD3 on MCC to test whether directed exploration closes the gap that local entropy/noise cannot.

7. Headline numbers

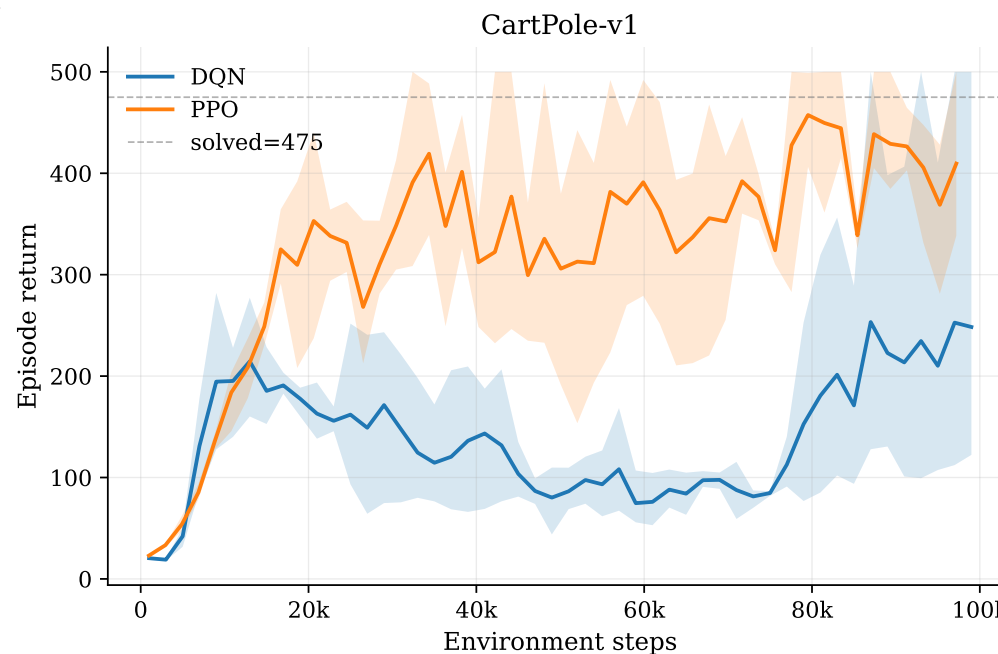
13×

SAC sample-efficiency over PPO on Pendulum (steps to -200)

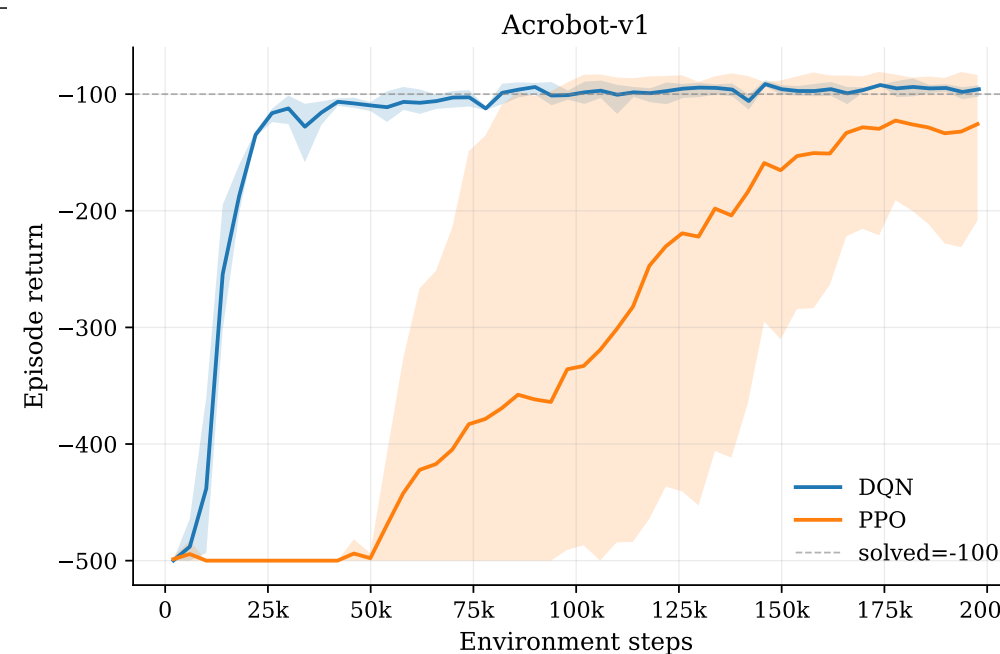
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Goal discoveries by SAC across α -sweep seeds on MCC

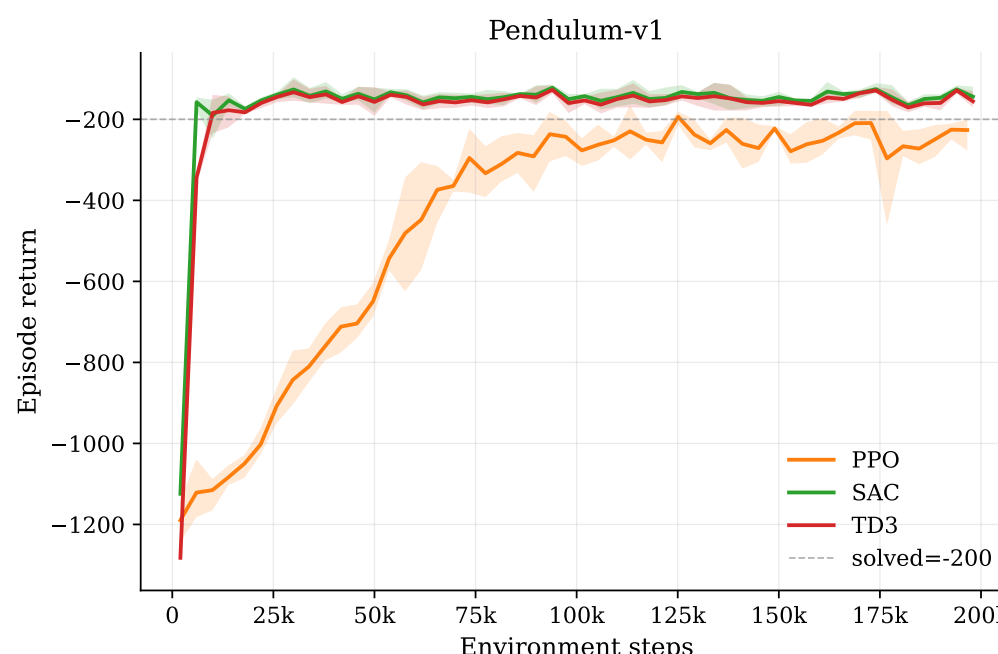
8. Learning curves: one figure per environment



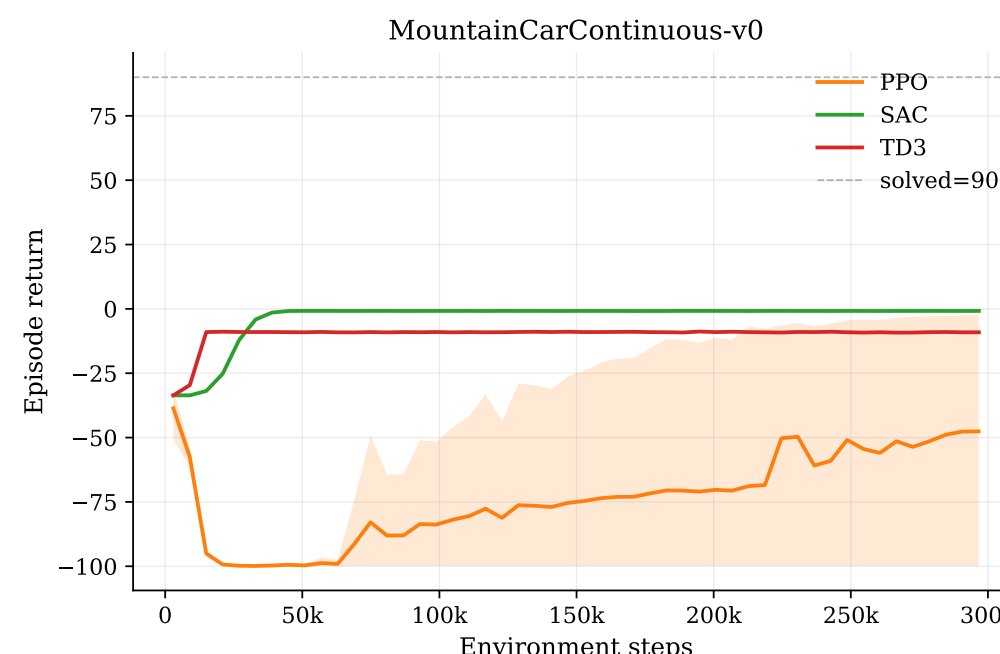
CartPole. **DQN bimodal** (1/3 solve); PPO reliable, 380, CI [350, 402].



Acrobot. DQN solves cleanly in $\sim 30k$ steps; one PPO seed collapses.

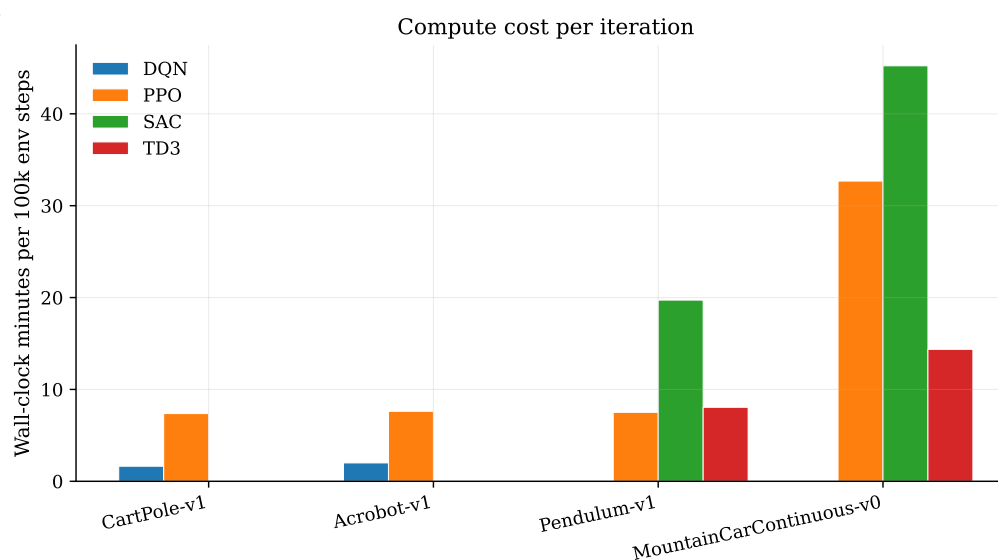


Pendulum. **SAC and TD3: $\sim 13\times$ more sample-efficient than PPO** (steps to -200).

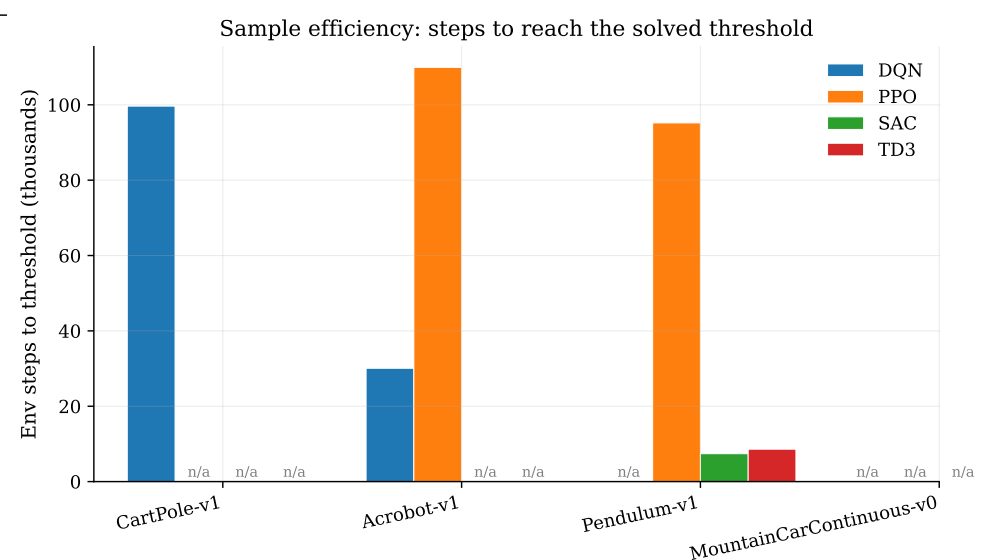


MCC, sparse. **Only PPO finds the goal (1/3 seeds only, -2.5)**; SAC and TD3 plateau at a lazy local optimum.

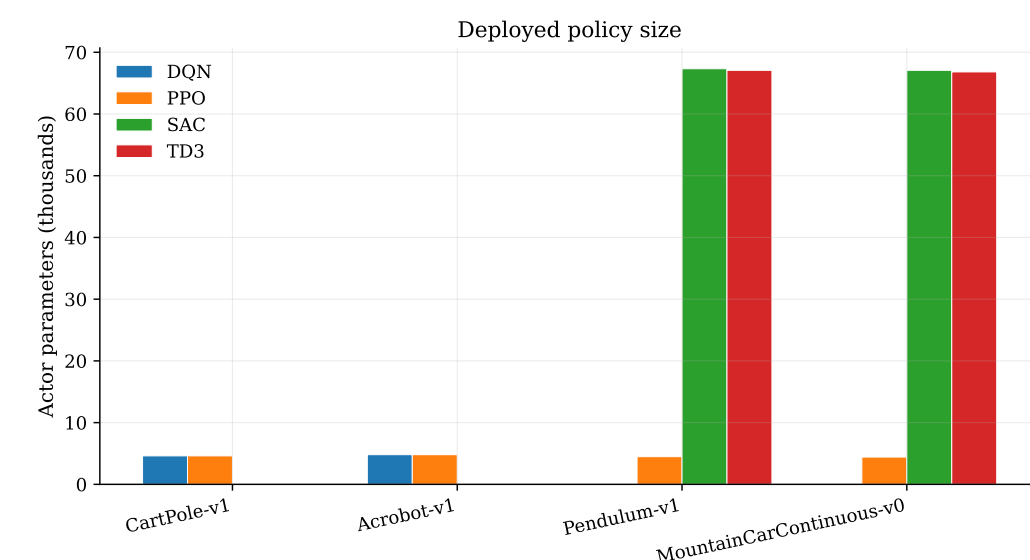
9. Cost, sample efficiency, model size



Compute: SAC > TD3 > PPO > DQN.

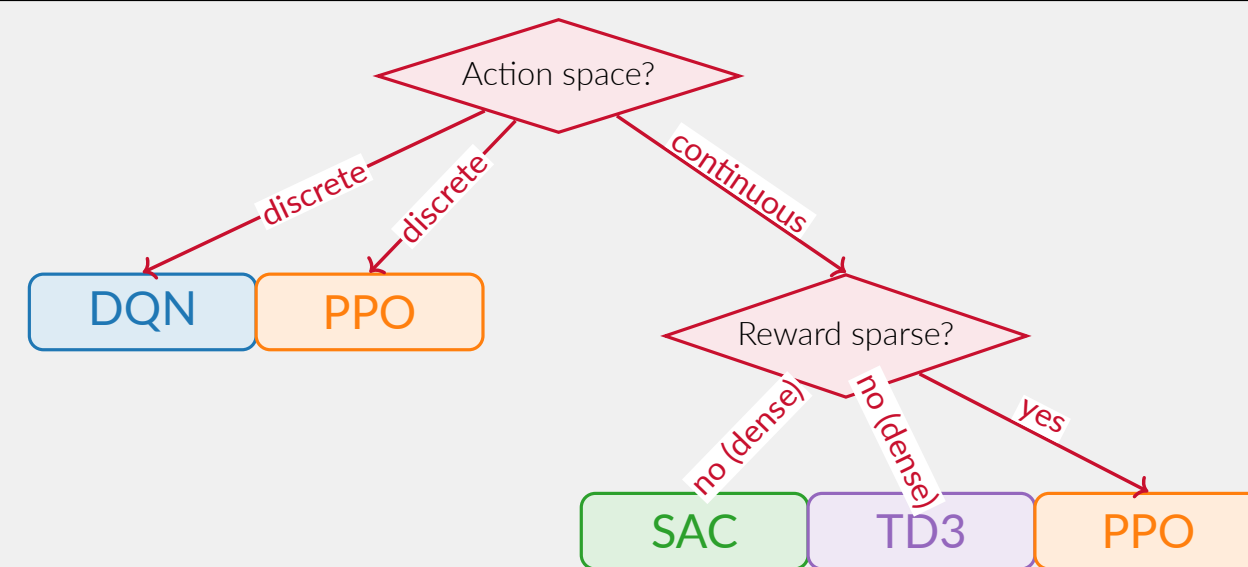


Sample efficiency (steps to threshold).



Actor params, (64, 64) vs (256, 256).

10. When to use which algorithm



- **Discrete actions:** **DQN** fastest on Acrobot ($\sim 30k$ steps); **PPO** more reliable on CartPole (avoids bimodality).
- **Dense-reward continuous:** **SAC** and **TD3** dominate, $12.9\times$ over **PPO** on Pendulum (steps to threshold).
- **Sparse-reward continuous:** **PPO** is the only one to find the MCC goal (1/3 seeds); the failure of SAC/TD3 is structural in the canonical per-step exploration mechanism, not a tuning issue.

Pick SAC or TD3 first on continuous control; switch to PPO (or add explicit exploration bonuses) the moment reward becomes sparse.

References & project artefacts

- [1] Mnih et al. *Playing Atari with Deep RL*. arXiv:1312.5602, 2013. [4] Fujimoto et al. *Addressing Function Approximation Error*. ICML, 2018.
- [2] Schulman et al. *PPO Algorithms*. arXiv:1707.06347, 2017. [5] Towers et al. *Gymnasium*. 2023.
- [3] Haarnoja et al. *Soft Actor-Critic*. ICML, 2018. [6] Raffin et al. *Smooth Exploration for Robotic RL*. CoRL, 2022.

Artefacts: PyTorch from scratch (no **stable-baselines3**); per-(algo, env) YAML in **configs/**; curves, seed grids, ablations and bar charts under **plots/**; bootstrap-CI scoreboard in **plots/scoreboard.{csv, tex}**.